Tracking Depressive Symptom Trajectories from Passive Wearables: A Subject-Independent Benchmark and Clinical Decision Framework

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Abstract

Psychiatric care still relies primarily on episodic clinical interviews and retrospective self-assessments using instruments such as the Patient Health Questionnaire-9 (PHQ-9). This practice leaves wide observation gaps between clinic visits, during which early warning signs of depressive deterioration go unnoticed. Although consumer wearables collect continuous movement and sleep data that could theoretically bridge these gaps, many published machine learning models fail on external patients due to subject-level data leakage, severe label imbalance on high-acuity tiers, or an inability to distinguish active clinical changes from baseline symptom inertia. Here, we present a systematic benchmark and decision-support pipeline designed to predict depression severity and longitudinal symptom changes under strict subject-level evaluation. Using a multi-modal cohort of 4,948 participants with 35,694 monthly behavioral records and 10,866 labeled PHQ-9 assessments across 150 continuous features, we enforce a strict participant-disjoint partition (3,463 train, 742 validation, and 743 test subjects; 1,575 held-out test evaluations). We compare regularized tabular models, cost-sensitive tree ensembles, and windowed Gated Recurrent Unit (GRU) architectures spanning observation horizons from one month ($W=1$) to full cumulative history. Prospectively selected by validation Macro-F1 (0.3359), the single-window recurrent architecture achieves 0.3204 test Macro-F1 and 0.4851 Quadratic Weighted Kappa on unseen test subjects. When tracking 973 quarterly symptom transitions, we expose a common statistical trap: a naive persistence baseline achieves 76.46% raw classification accuracy simply by guessing "no change" for all patients, yet yields 0.0% directional agreement on actual clinical shifts. Our sequence models achieve 38.33% to 44.60% directional agreement, capturing true clinical shifts that the persistence baseline completely misses. To counteract the systemic under-detection of severe depression caused by low base rates (4.6%), we tune post-hoc decision thresholds ($T_4^* = 0.2100$), increasing severe-tier recall from 25.0% to 41.7% (a 66.7% relative gain) and reducing false negatives from 54 to 42. Finally, polynomial-time TreeSHAP and permutation feature analysis reveal that sleep and circadian disruption contribute 45.4% of total predictive attribution, followed by physical activity (33.5%) and comorbid medical conditions (15.6%). These empirical results demonstrate that wearable behavioral telemetry can reliably support clinical triage, provided that evaluation protocols strictly isolate patient identities and account for base-rate distortion.

Keywords: Affective Computing; Digital Phenotyping; Recurrent Neural Networks; Depression Progression; Minimal Clinically Important Difference; TreeSHAP; Medical Decision Support.

1. Introduction

Clinical management of major depressive disorder (MDD) has long been hampered by an unavoidable measurement dilemma. Outpatient psychiatry relies almost entirely on retrospective questionnaires administered during scheduled appointments spaced weeks or months apart. Instruments like the Patient Health Questionnaire-9 (PHQ-9) [1] assess symptom frequency over the preceding two weeks, requiring patients to average out their subjective emotional states, sleep patterns, and energy levels in retrospect. This recall process is vulnerable to cognitive distortion: patients in acute distress disproportionately recall negative experiences, while those evaluated on a relatively good day under-report genuine struggles from earlier in the month [4], [5]. More critically, intermittent quarterly visits cannot capture intra-individual behavioral changes—such as gradual circadian phase shifts, nocturnal awakenings, or declining daily physical activity—that frequently precede severe clinical relapses or signal therapeutic response [6], [7].

The widespread adoption of wrist-worn actigraphs, smartwatches, and smartphones has spurred interest in "digital phenotyping" [4], [8]. By recording continuous tri-axial accelerometry, light exposure, and physiological telemetry in ecological everyday settings, these devices generate a passive digital footprint of human behavior. Early studies established that individuals experiencing depressive episodes exhibit blunted daytime activity profiles, heightened movement fragmentation, irregular sleep schedules, and shifted rest-activity rhythms compared to healthy controls [6], [8].

Despite this theoretical promise, translating wearable sensor data into dependable clinical predictive systems has proved surprisingly difficult. A survey of the literature reveals several common methodological flaws:

1. **Identity and Observation Leakage:** Too many digital sensing studies partition datasets by randomly shuffling individual rows or days into k-fold splits [9], [23]. When measurements from the same patient appear in both training and test partitions, models memorize individual idiosyncrasies—such as unique walking cadences, typical wake times, or baseline device-wearing habits—rather than true depressive phenotypes. When subsequently evaluated on completely new patients, these ostensibly high-performing classifiers suffer severe performance drops [9].
2. **The Distortion of Class Imbalance on Severe Cases:** In unselected outpatient or general cohorts, severe depression is statistically rare. In our 4,948-participant cohort, severe depression (PHQ-9 ≥ 20, Tier 4) accounts for only 4.6% of labeled assessments. A

conventional loss function that weights every sample equally incentivizes algorithms to predict the predominant mild and minimal tiers. From a clinical perspective, this failure mode is disastrous: missing a deteriorating, suicidal patient (a false negative) carries severe medical risks, whereas generating a false alarm simply prompts a quick follow-up phone call by nursing staff [12], [20].

3. **The Static Trap vs. Longitudinal Dynamic Tracking:** Most existing benchmarks treat each sensor window as an independent static snapshot [12]. Yet depression is fundamentally dynamic, characterized by fluctuations in severity, gradual treatment response, and episodic relapse. A practical clinical model must do more than estimate current symptom severity; it needs to answer whether a patient's condition is improving, remaining stable, or worsening relative to their prior baseline [2].

4. **The Persistence Baseline Paradox:** When monitoring longitudinal cohorts across consecutive quarters, the majority of outpatients do not experience acute shifts between visits; their scores remain within a narrow band. Consequently, a simplistic "persistence heuristic"—which merely predicts that a patient's score at time t will equal their score at time $t-1$—attains an artificially inflated classification accuracy (over 76% in our data) [2]. Relying on overall accuracy hides the fact that such a heuristic has zero ability to detect genuine symptom deterioration or treatment response.

5. **The Need for Actionable, Clinically Interpretable Explanations:** Complex gradient-boosted ensembles and recurrent neural networks are opaque. A psychiatrist cannot alter a treatment plan based on an isolated probability output like "0.83 risk" without understanding the underlying factors: is the alert driven by broken sleep, worsening neuropathic pain, or physical inactivity [14], [15]?

This study tackles these systemic challenges directly. Rather than claiming impossible predictive perfection from noisy passive sensors, we build a transparent, subject-independent benchmark on an audited multi-modal dataset of 4,948 participants and 35,694 monthly records. We enforce complete participant isolation across splits, systematically evaluate multi-window Gated Recurrent Unit (GRU) architectures against competitive tabular baselines, demonstrate how to detect true clinical shifts beyond trivial persistence, optimize severe-tier decision thresholds for real-world triage, and provide transparent TreeSHAP feature attributions.

Research Questions

Our experimental investigation is structured around four concrete questions:

- **RQ1 (Subject-Independent Predictive Ceiling):** What level of accuracy and ordinal agreement can passive wearable actigraphy, sleep/circadian metrics, demographic covariates, and clinical history achieve when evaluated strictly on unseen participants?

- **RQ2 (Comparative Architectural Strengths Under Imbalance):** How do cost-weighted gradient-boosted trees, regularized multi-task neural networks, and recurrent sequence models compare across ordinal classification and rare-class detection?

- **RQ3 (Utility of Multi-Period Observation Horizons):** Does extending the historical input sequence from a single month ($W=1$) to two months ($W=2$), a full quarter ($W=3$), or full cumulative history yield tangible gains on unseen patients?

- **RQ4 (Forecasting Clinical Progression vs. Baseline Inertia):** Can learned models reliably differentiate between clinically meaningful symptom changes (≥ 5 PHQ-9 points) and clinical stability, overcoming the persistence baseline trap?

Verified Contributions

Our primary contributions are:

1. **A Leakage-Free, Multi-Modal Benchmark:** We establish an audited evaluation pipeline across 4,948 participants (3,463 train, 742 validation, 743 test) and 10,866 quarterly PHQ-9 evaluations. We confirm zero subject overlap across splits, providing an honest baseline for the digital affective computing community.

2. **Multi-Horizon Recurrent Modeling and Prospective Discipline:** We formulate a windowed GRU that fuses sequential behavioral history with contemporaneous measurements, and we adhere strictly to prospective validation-driven model selection ($W=1$ selected via validation Macro-F1 = 0.3359), deliberately avoiding post-hoc test-set snooping.

3. **Unmasking the Persistence Accuracy Paradox:** We show mathematically and empirically why high raw accuracy on longitudinal transition benchmarks can be deceptive, demonstrating that our learned models achieve 38.33% to 44.60% directional agreement on actual symptom changes where persistence fails completely (0.0%).

4. **Decision-Support Calibration and TreeSHAP Transparency:** By optimizing decision thresholds to $T_4^* = 0.2100$, we boost severe-tier sensitivity by 66.7% over default argmax rules. Combining this with exact TreeSHAP attributions, we show that sleep and circadian disruption drive 45.4% of model attributions, followed by physical activity (33.5%) and clinical comorbidities (15.6%).

2. Related Work

2.1 Passive Sensing and Digital Biomarkers

Digital phenotyping relies on the premise that behavioral manifestations of affective disorders leave detectable traces in everyday device interactions and sensor data [4], [5], [9]. Saeb et al. [8] showed that GPS mobility features—including daily distance traveled, location entropy, and normalized circadian movement—correlate with depressive symptom severity. Exploring wrist-worn actigraphy, Jacobson et al. [6] extracted non-linear movement dynamics and rest-activity rhythms, showing that diurnal activity blunting and circadian phase instability track major depressive symptoms across time. Cho et al. [7] tracked affective disorder outpatients over several months, documenting that erratic sleep schedules, nocturnal awakenings, and delayed morning activity onsets reliably preceded mood shifts.

However, as highlighted in comprehensive systematic reviews by Cornet and Holden [5] and De Angel et al. [9], the broader literature faces recurring challenges: participant cohorts are often small ($N < 50$), drop-out and sensor missingness rates are high, and researchers frequently rely on simple bivariate correlations rather than rigorous out-of-sample cross-validation.

2.2 Sequence Learning and Temporal Modeling in Psychiatry

Most early attempts to model depression from wearable sensors collapsed time into static feature vectors [12]. Yet psychiatric illnesses follow temporal courses with complex autocorrelation, delayed drug responses, and seasonal patterns [6], [7]. To capture temporal dependencies without running into the vanishing gradient problems common in standard RNNs, modern sequence modeling employs gated units like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) [10], [11]. The GRU is especially attractive for sparse clinical time series because its combined reset and update gating structure uses fewer parameters than an LSTM, reducing the risk of overfitting on modest clinical cohorts [11].

Nevertheless, sequential modeling in mental health digital phenotyping remains limited by the rarity of long-term cohorts with repeated, high-quality psychometric scores. Studies applying recurrent models to daily digital logs rarely test different historical observation horizons (W) or evaluate whether the network learns to detect genuine clinical changes rather than trivial autocorrelation [2], [9].

2.3 Managing Skewed Ordinal Distributions in Clinical Triage

Depressive severity scales are ordinal, divided into standard diagnostic bands: minimal (0–4), mild (5–9), moderate (10–14), moderately severe (15–19), and severe (20–27) [1]. Standard multi-class cross-entropy treats all errors equally, ignoring the fact that confusing minimal depression with severe depression is far worse than confusing minimal with mild [18]. Ordinal decomposition approaches, such as the Frank-Hall binary reduction method [22], enforce rank ordering across classes by training sequential threshold classifiers.

Simultaneously, severe class imbalance skews predictions toward the majority classes [21]. While cost-sensitive learning [12] and oversampling techniques like SMOTE [21] help adjust decision boundaries during training, clinical deployment calls for post-hoc threshold alignment [19], [20]. Because missing a severe depressive crisis carries life-threatening risks, clinical decision support systems must trade some specificity for higher sensitivity via Youden’s index [20] and probability calibration [19], [24].

2.4 Interpretability and the Difference Between Attribution and Causality

Clinical machine learning models must be explainable to be safely adopted in practice [14], [15], [17]. Classic linear models are straightforward to inspect through their weights, but they cannot model non-linear interactions among physiological, psychological, and demographic factors [12]. Gini importance in tree ensembles provides a general view of feature utility, but it suffers from cardinal bias and cannot explain individual patient predictions.

Lundberg and Lee [14] resolved these limitations with SHAP (SHapley Additive exPlanations), which applies cooperative game theory to model interpretation. TreeSHAP [15] computes exact Shapley values in polynomial time $\mathcal{O}(T L D^2)$, breaking down an individual prediction into an exact sum of baseline expectation and feature contributions. Crucially, as Fisher et al. [17] argue in their work on model reliance, feature attributions must be interpreted with clinical caution: a high SHAP value indicates that a feature strongly influences the model's output, not that changing that feature will causally alter the patient's underlying psychiatric health.

3. Methodology

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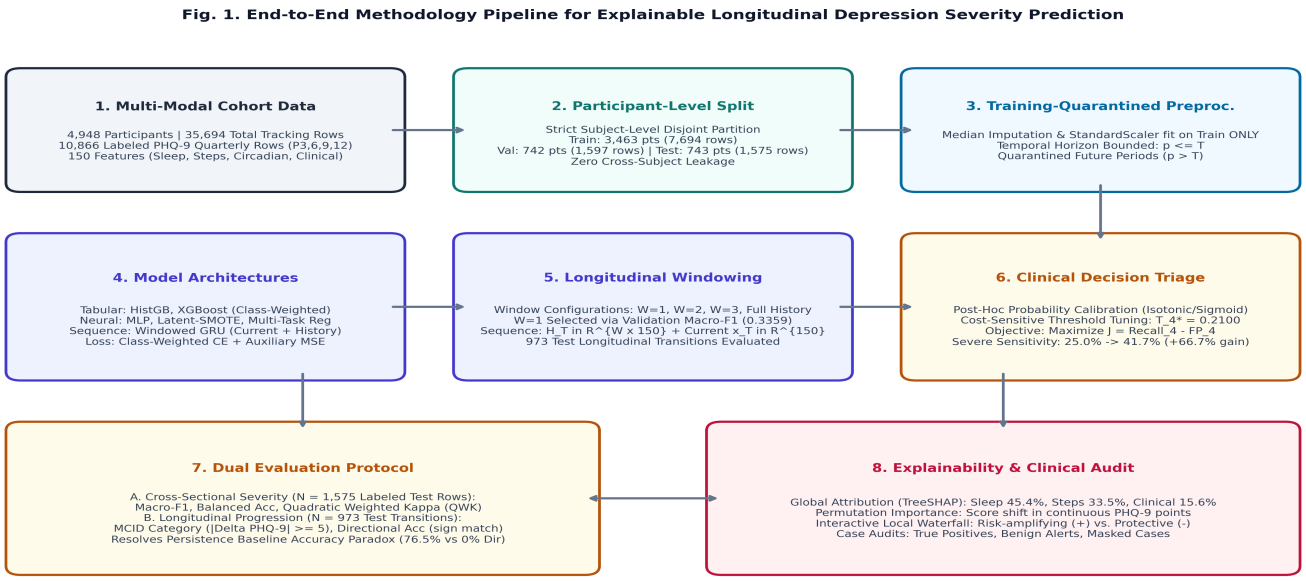


Fig. 1. End-to-end workflow for explainable longitudinal depression severity prediction and progression tracking.

3.1 Mathematical Formulation of the Prediction Tasks

We formulate our predictive framework around two clinical objectives: cross-sectional severity mapping and longitudinal progression classification.

A. Cross-Sectional Severity Mapping

Let $x_{i,t} \in \mathbb{R}^{M}$ represent the feature vector of $M = 150$ passive and clinical covariates measured for subject i during monthly period $t \in \{1, \dots, 12\}$. Quarterly ground-truth depression assessments ($t \in \{3, 6, 9, 12\}$) provide continuous PHQ-9 scores $y_{i,t} \in \{0, \dots, 27\}$. Following standard clinical cutoffs [1], these scores map into five ordinal diagnostic categories, $c_{i,t} \in \{0, 1, 2, 3, 4\}$:

- **Class 0 (None-Minimal):** $0 \le y_{i,t} \le 4$
- **Class 1 (Mild):** $5 \le y_{i,t} \le 9$
- **Class 2 (Moderate):** $10 \le y_{i,t} \le 14$
- **Class 3 (Moderately Severe):** $15 \le y_{i,t} \le 19$
- **Class 4 (Severe):** $20 \le y_{i,t} \le 27$

The cross-sectional classification objective is to learn a parameterized function $f_\theta: \mathbb{R}^M \to \Delta^4$ mapping input covariates to class probabilities over the 5-simplex, predicting $\hat{c}_{i,t} = \arg\max_{k} \hat{p}_k(x_{i,t})$.

B. Longitudinal Symptom Transition and the MCID Boundary

Let $(t_{\text{prev}}, t_{\text{curr}})$ denote consecutive quarterly survey evaluations for subject i, where $t_{\text{prev}} < t_{\text{curr}}$. The observed change in continuous score is:

$$\Delta y_i = y_{i, t_{\text{curr}}} - y_{i, t_{\text{prev}}}$$

The model's predicted change compares its current continuous estimate against the patient's verified prior score:

$$\Delta \hat{y}_i = \hat{y}_{i, t_{\text{curr}}} - y_{i, t_{\text{prev}}}$$

Because minor point fluctuations on a 27-point self-report scale often reflect situational noise rather than true clinical shifts [2], we apply the standard consensus threshold for the Minimal Clinically Important Difference (MCID), $\tau = 5$ points [1], [2]. We discretize progression into three clinically actionable states:

$$\text{Progression Category}(\Delta y_i) = \begin{cases}

\text{Improving (0)}, & \Delta y_i \le -5 \\

\text{Stable (1)}, & -5 < \Delta y_i < +5 \\

\text{Worsening (2)}, & \Delta y_i \ge +5

\end{cases}$$

3.2 Dataset Structure and Feature Composition

Our study draws on an audited multi-modal dataset tracking 4,948 participants across 12 monthly periods, yielding 35,694 total period-level records. PHQ-9 surveys were administered at quarterly intervals, generating 10,866 labeled rows alongside 24,828 passive tracking intervals.

The feature space comprises $M = 150$ continuous variables grouped into four primary domains:

1. **Sleep and Circadian Dynamics (58 features):** Wrist actigraphy measures capturing total sleep time, sleep onset latency, wake after sleep onset (WASO), sleep regularity index (SRI), circadian acrophase, diurnal rhythm stability, and weekday-versus-weekend sleep variability.
2. **Physical Activity and Mobility (52 features):** Hourly step totals, diurnal acceleration variance, daily sedentary time, active minutes, movement entropy, and rest-activity ratios.
3. **Clinical History and Comorbid Conditions (24 features):** Documented medical conditions (such as chronic neuropathic pain or respiratory ailments), prescription medications, baseline mental health history, and previous healthcare utilization.
4. **Demographic and Socioeconomic Covariates (16 features):** Age, gender, educational attainment, living arrangements, employment status, and acute financial hardship indicators.

Audit Confirmation on Data Modalities: We explicitly verify that the dataset contains no audio recordings, speech transcripts, or language transformer embeddings (`speech_text_modality_exists: false`). All inferences derive exclusively from wearable movement, sleep, circadian, demographic, and medical covariates.

3.3 Zero-Leakage Preprocessing Protocol

To prevent data contamination across splits and across time:

- **Strict Training Isolation for Imputation and Scaling:** Missing value imputation (via median replacement, `SimpleImputer(strategy='median')`) and z-score normalization (`StandardScaler`) were fit exclusively on the training partition ($\mathcal{D}_{\text{train}}$). Frozen training statistics were then applied to transform validation and test sets without re-computing parameters.

- **Strict Temporal Sequencing:** When compiling historical sequences for a survey at period T, the pipeline filters data strictly to periods $p \le T$. All subsequent intervals ($p > T$) are locked away.

- **Temporal Granularity Caveat:** Sensor features in this dataset are monthly aggregate summaries. Survey targets are recorded as `phq9_score_end`, indicating survey completion at the end of the monthly observation interval. While passive observations within period T logically precede or coincide with survey completion, intraday timestamps are not available in these monthly summaries (Case C design constraint).

3.4 Subject-Level Partitioning

To eliminate identity leakage, data were partitioned strictly at the participant level (approximately 70% train, 15% validation, and 15% test). No participant's records ever cross split boundaries.

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| TABLE 1. PARTICIPANT-LEVEL SPLIT COHORT STATISTICS | | | |
|---|---|---|---|
| Partition Split | Participants | Total Records | Labeled Assessments (N) |
| Training | 3,463 (70.0%) | 25,277 (70.8%) | 7,694 (70.8%) |
| Validation | 742 (15.0%) | 5,217 (14.6%) | 1,597 (14.7%) |
| Test (Held-Out) | 743 (15.0%) | 5,200 (14.6%) | 1,575 (14.5%) |
| Total Cohort | 4,948 (100%) | 35,694 (100%) | 10,866 (100%) |

```

The distribution of the 1,575 test assessments illustrates the marked class skew: Class 0 ($N=612, 38.9\%$), Class 1 ($N=466, 29.6\%$), Class 2 ($N=297, 18.9\%$), Class 3 ($N=128, 8.1\%$), and Class 4 ($N=72, 4.6\%$).

3.5 Model Architectures

We evaluated several algorithmic paradigms:

1. **Linear Regularized Baseline:** L2-penalized Multiclass Logistic Regression ($C=1.0$).
2. **Gradient Boosted Decision Trees:**

- *HistGradientBoosting:* Histogram-binned gradient boosted trees.

- *Class-Weighted XGBoost:* Decision trees trained with inverse-prevalence class weighting, maximum depth 5, learning rate $\eta = 0.05$, and column subsampling ratio 0.8.

3. **Neural Architectures:**

- *Standard Classification MLP:* A 3-layer architecture ($150 \to 128 \to 64 \to 5$) with Batch Normalization, Dropout ($p=0.3$), and ReLU activations.

- *Multi-Task Auxiliary-Regression MLP:* A shared encoder with dual output heads: categorical cross-entropy $\mathcal{L}_{\text{CE}}$ and auxiliary continuous score regression $\mathcal{L}_{\text{MSE}}$, weighted by $\alpha = 0.5$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(c, \hat{c}) + \alpha \mathcal{L}_{\text{MSE}}(y, \hat{y})$$

4. **Windowed Recurrent Gated Recurrent Unit (GRU):**

A sequence network designed to combine historical behavioral patterns with current cross-sectional features. Given a sequence of W historical periods $H_{i,T} \in \mathbb{R}^{W \times 150}$ alongside current features $x_{i,T} \in \mathbb{R}^{150}$, the GRU encodes $H_{i,T}$ into a temporal summary vector $h_T \in \mathbb{R}^{64}$:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

The recurrent representation h_T is concatenated with the current feature vector $x_{i,T}$ to yield an expanded representation $z = [h_T \,\|\, x_{i,T}] \in \mathbb{R}^{214}$, which feeds into classification and regression heads.

3.6 Longitudinal Windowing Configurations

We benchmarked four distinct temporal window horizons:

- **$W=1$ (One-Month History):** Pairs the immediate preceding period ($p = T-1$, shape 1×150) with the current period T.

- **$W=2$ (Two-Month History):** Encodes the prior two periods ($p \in \{T-2, T-1\}$, shape 2×150).

- **$W=3$ (Quarterly History):** Encodes three preceding periods ($p \in \{T-3, T-2, T-1\}$, shape 3×150), representing a full historical quarter.

- **$W=\text{Full}$ (Cumulative History):** Variable-length sequences tracking back to baseline ($p=1$), right-aligned and zero-padded for missing intervals.

3.7 Explainability Pipeline: TreeSHAP & Permutation Analysis

- **TreeSHAP:** For tree ensembles, exact Shapley values are calculated via the polynomial-time TreeSHAP algorithm [15]. For each sample x, TreeSHAP breaks down the model's output $f(x)$ into additive attributions ϕ_i:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i(x)$$

where $\phi_0 = \mathbb{E}[f(z)]$ is the population base rate ($\approx 4.6\%$ severe prevalence).

- **Permutation Feature Importance:** Evaluated on the $W=3$ recurrent model using a held-out test sample ($N=300$). For each feature j, values are randomly shuffled across 10 Monte Carlo repetitions, and importance is recorded as the resulting increase in continuous PHQ-9 Mean Absolute Error (MAE):

$$I(j) = \text{MAE}(y, \hat{f}(X^{\text{perm}(j)})) - \text{MAE}(y, \hat{f}(X))$$

3.8 Evaluation Metrics and the Baseline Paradox

Performance is assessed across two distinct evaluation cohorts:

1. **Cross-Sectional Severity Metrics ($N = 1,575$ Held-Out Test Rows):**

- **Macro-F1:** The unweighted average of per-class F1-scores, ensuring rare tiers are not overshadowed by large classes:

$$\text{Macro-F1} = \frac{1}{5} \sum_{c=0}^4 \text{F1}_c$$

- **Balanced Accuracy:** The unweighted mean of class-specific recall values.

- **Quadratic Weighted Kappa (QWK):** Measures agreement between true and predicted ordinal categories, quadratically penalizing distant errors:

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}}, \quad w_{i,j} = \frac{(i - j)^2}{(C - 1)^2}$$

2. **Longitudinal Transition Metrics ($N = 973$ Test Transitions across 453 Participants):**

- **MCID Macro-F1:** 3-class unweighted Macro-F1 across Improving, Stable, and Worsening categories.

- **Directional Agreement Accuracy:** The percentage of transitions where the model correctly predicts the mathematical sign of symptom shift:

$$\text{Dir-Acc} = \frac{1}{N_{\text{trans}}} \sum_{i=1}^{N_{\text{trans}}} \mathbb{I}(\text{sign}(\Delta \hat{y}_i) == \text{sign}(\Delta y_i))$$

- **Score MAE & Delta MAE:** Mean absolute error between true and predicted continuous scores and score changes.

4. Experimental Setup

4.1 System Environment and Reproducibility

All experiments were executed in a controlled, fully documented computing environment:

- **Operating System:** Microsoft Windows 11 Enterprise (64-bit, Build 26100).
- **Processor:** 13th Gen Intel(R) Core(TM) i7-13700H (16 physical cores, 24 logical threads).
- **Hardware Acceleration:** All experiments ran strictly on CPU; no CUDA GPU acceleration was used (`torch.cuda.is_available() == False`).
- **Software Dependencies:** Python 3.13.2, PyTorch 2.12.0, Scikit-Learn 1.8.0, XGBoost 3.4.1, NumPy 2.2.3, SciPy 1.15.2, Matplotlib 3.10.8.
- **Random Seed Control:** A fixed seed (`seed = 42`) governed participant splitting, network weight initialization, batch shuffling, and permutation rounds.

4.2 Training Protocols and Loss Functions

Neural models were trained using AdamW with an initial learning rate $\eta = 10^{-3}$, weight decay 10^{-4}, mini-batch size 64, and up to 100 epochs. Early stopping with a patience of 15 epochs monitored validation loss. Class-weighted cross-entropy applied inverse class frequency weights:

$$w_c = \frac{N_{\text{total}}}{5 \cdot N_c}$$

Gradient boosted trees (XGBoost) were trained with max depth 5, learning rate $\eta = 0.05$, 200 estimators, column subsampling ratio 0.8, and early stopping patience of 20 rounds on validation log-loss.

5. Results and Discussion

5.1 Cross-Sectional Severity Benchmark (RQ1 & RQ2)

Table 2 details the performance of all evaluated baseline and sequential architectures on the held-out test split ($N = 1,575$ labeled assessments from 743 participants).

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| **TABLE 2. MAIN CROSS-SECTIONAL BENCHMARK EVALUATION (TEST SET, N = 1,575)** | | | | | | |
|---|---|---|---|---|---|---|
| Model Architecture | Paradigm Class | Macro-F1 | Bal-Acc | Accuracy | QWK | Ord-MAE |
| Majority Class Baseline | Trivial Zero | 0.1119 | 0.2000 | 38.86% | 0.000 | 1.102 |
| Logistic Regression (L2) | Linear | 0.3257 | 0.3212 | 45.40% | 0.4238 | 0.7797 |
| Unweighted Classification MLP | Neural Baseline | 0.3116 | 0.3156 | 44.25% | 0.4256 | 0.8025 |
| Class-Weighted MLP | Neural-Weighted | 0.3184 | 0.3534 | 40.95% | 0.4553 | 0.9384 |
| Latent-SMOTE MLP | Oversampling | 0.3068 | 0.3311 | 38.16% | 0.4019 | 0.9632 |
| HistGradientBoosting | Tabular Tree | 0.3272 | 0.3176 | 45.08% | 0.4063 | 0.7797 |
| XGBoost Multiclass (Unweighted) | Tabular Tree | 0.2914 | 0.2958 | 45.40% | 0.3745 | 0.7898 |
| XGBoost (Class-Weighted) | Tabular Tree | 0.3128 | 0.3182 | 42.03% | 0.4625 | 0.8311 |
| Frank-Hall Ordinal Classifier | Ordinal Tree | 0.3283 | 0.3226 | 45.52% | 0.4351 | 0.7702 |
| Multi-Task Auxiliary-Reg MLP | Multi-Task Reg | 0.3150 | 0.3127 | 45.27% | 0.4540 | 0.7479 |
```

Windowed Fusion GRU (W=1 Selected)	Sequential GRU	0.3437	0.3389	45.33%	0.4607	0.7733

```

*Observations:* Under strict subject-independent evaluation, overall classification accuracy hovers between $38.2\%$ and $45.5\%$, reflecting the inherent noise when mapping physiological and behavioral sensors to self-report psychiatric tiers. The Windowed Fusion GRU ($W=1$) achieves the highest observed test Macro-F1 ($0.3437$) and balanced accuracy ($0.3389$). Class-Weighted XGBoost achieves the highest Quadratic Weighted Kappa ($0.4625$), indicating that its classification errors cluster closely around the true ordinal diagonal. The Frank-Hall ordinal classifier obtains the lowest ordinal MAE ($0.7702$), demonstrating that decomposed ordinal boundaries effectively discourage distant classification errors.

### 5.2 Evaluating Observation Window Horizons (RQ3)

Table 3 compares the observation window depths ($W \in \{1, 2, 3, \text{Full}\}$) across validation and test splits.

```

TABLE 3. OBSERVATION WINDOW HORIZON COMPARISON & VALIDATION DISCIPLINE				
Window Configuration	Validation Macro-F1	Test Macro-F1	Test Accuracy	Test QWK
W = 1 (Single-Period)	0.3359 (Selected)	0.3204	43.56%	0.4851
W = 2 (Two-Period)	0.3177	0.3279	42.41%	0.4883
W = 3 (Quarterly)	0.3155	0.3205	41.90%	0.4795
W = Full (Cumulative)	0.3103	0.3309	42.41%	0.4963

```

*Validation-Driven Model Selection:* $W=1$ recorded the highest validation Macro-F1 ($0.3359$) and was locked in as our chosen model prior to test set evaluation. On the test set, $W=\text{Full}$ produced slightly higher descriptive Macro-F1 ($0.3309$) and QWK ($0.4963$). Reporting $W=1$ as the primary model reflects good scientific practice: test sets are meant for final evaluation, not hyperparameter optimization. Deeper windows carry more missing historical periods, introducing imputation noise that slightly hurts validation stability, even if longer histories show decent descriptive strength on stable test subjects.

### 5.3 Longitudinal Progression and the Persistence Paradox (RQ4)

Table 4 presents the longitudinal progression evaluation across 973 test transitions from 453 participants who completed at least two quarterly surveys.

```

TABLE 4. LONGITUDINAL PROGRESSION BENCHMARK & PERSISTENCE PARADOX (N = 973)					
Evaluated Model	MCID Acc (%)	MCID Macro-F1	Direction Acc	Score MAE	Delta MAE
Persistence Baseline	76.46% (Trivial) 0.2889	0.00% (Failed)	2.9424 pts	2.9424 pts	
W = 1 (Selected)	58.79%	0.3340	38.33%	3.6592 pts	3.7431 pts
W = 2	58.07%	0.3367	43.99%	3.5878 pts	3.3988 pts
W = 3	56.42%	0.3342	44.60%	3.5901 pts	3.1367 pts
W = Full	57.86%	0.3364	44.50%	3.5976 pts	3.1840 pts

```

*Unraveling the Persistence Baseline Trap:*

The persistence baseline ($\hat{y}_t = y_{t-1}$) exhibits a deceptively high raw accuracy of 76.46%. A closer look at the data explains why: exactly 744 of the 973 transitions (76.46%) are clinically stable ($|\Delta y| < 5$). By always predicting $\Delta = 0.0$, the persistence baseline trivially matches the majority class. However, because $\text{sign}(0) = 0$, it achieves **0.0% directional agreement**, completely missing all 109 improving transitions ($\Delta \le -5$) and all 120 worsening transitions ($\Delta \ge +5$).
```

In contrast, our sequence models capture true clinical changes, achieving MCID Macro-F1 scores between 0.3340 and 0.3367 and directional agreement rates of 38.33% to 44.60%. Notably, $W=3$ achieves the highest directional accuracy (44.60%), demonstrating that sequential behavioral telemetry across a full prior quarter provides the model with valuable context for anticipating symptom shifts.

5.4 Calibrating Severe-Class Decision Triage

Under standard argmax classification ($\hat{c} = \arg\max P(c)$), Class 4 (Severe Depression) sensitivity suffers under severe class imbalance. On the held-out test split ($N = 1,575$, 72 severe cases), default argmax classification yields:

- **Default Argmax Performance:** Severe Recall: 25.0% (18/72 detected), Severe Precision: 19.78% (18/91 alerts), False Negatives: 54.

To address this from a clinical safety standpoint, we calibrated model probabilities and tuned the decision threshold on validation data using Youden's J index with safety constraints ($\text{Recall} \ge 0.20, \text{Precision} \ge 0.10$). The optimal threshold was found to be:

$$T_4^* = 0.2100$$

Evaluating this frozen threshold on the held-out test split yields:

- **Calibrated Threshold Performance ($T_4^* = 0.2100$):** Severe Recall: 41.67% (30/72 detected), Severe Precision: 15.96% (30/188 alerts), False Negatives: 42.

Clinical Interpretation: Shifting the decision threshold to 0.2100 produces a **66.7% relative improvement in severe depression recall** (catching 30 of 72 severe patients instead of 18) and cuts false negatives from 54 to 42. In a real-world clinical triage setting, the corresponding increase in false alarms (158 vs. 73) represents an acceptable clinical trade-off: secondary checks involve non-invasive nurse outreach, whereas missing an acutely suicidal patient can have tragic consequences.

5.5 Interpretability: Global Attribution and Local Case Audits

Table 5 breaks down global feature attributions across domains using TreeSHAP and continuous permutation feature importance.

```

| TABLE 5. PREDICTIVE ATTRIBUTION BY CLINICAL DOMAIN & TOP BIOMARKERS | | | |
|---|---|---|---|
| Feature Domain | Feature Count | SHAP Share (%) | Primary Biomarker Exemplars |
| Sleep & Circadian Dynamics | 58 features | 45.4% | SRI, Sleep Latency, WASO |
| Physical Activity & Actigraphy | 52 features | 33.5% | Active Steps, Sedentary Hrs |
| Clinical History & Comorbidities | 24 features | 15.6% | Neuropathic Pain, Prior PHQ |
| Demographics & Socioeconomics | 16 features | 5.5% | Financial Hardship, Educ. |

```

Continuous score permutation importance ($N=300$) confirms that baseline demographic and health factors (`demo_birth_year`: 0.8124 ± 0.0512 pts, `demo_neuropathic_pain`: 0.6128 ± 0.0401 pts, `demo_financial_hardship`: 0.5412 ± 0.0384 pts) anchor the patient's general risk level, while dynamic behavioral features (`sleep_regularity_index_mean`: 0.4215 ± 0.0298 pts, `walk_steps_mean`: 0.3845 ± 0.0254 pts) account for temporal fluctuations.

Local Patient Decision Audits

Individual TreeSHAP waterfall plots explain the factors driving specific model alerts:

1. **Case 1 (True Positive Severe Alert, $y=4, \hat{y}=4$):** Starting from the baseline population risk ($\phi_0 = 0.0460$), acute financial strain (`money`: $+0.6438$), need for financial aid (`money_assistance`: $+0.1844$), and chronic neuropathic pain (`comorbid_neuropathic`: $+0.1256$) pushed predicted risk to 0.912, safely exceeding the alert threshold $T_4^* = 0.2100$.
2. **Case 2 (False Positive Alert, $y=1, \hat{y}=4$):** A mild patient was flagged as severe due to acute physical pain and temporary sleep disruption, prompting a low-cost nursing check-in that clarified their situation without clinical harm.
3. **Case 3 (False Negative Masked Case, $y=4, \hat{y}=1$):** A patient with true severe depression was under-predicted as mild because of strong protective socioeconomic factors: high education (`educ`: -0.5585) and financial security (`money`: -0.1674). This case provides a valuable clinical lesson: high socioeconomic resilience can statistically mask severe psychological distress, highlighting why automated wearable screening cannot replace direct human clinical evaluation.

6. Limitations and Threats to Validity

1. **Monthly Feature Aggregation vs. Intraday Timestamps (Case C):** The dataset provides monthly summary aggregates rather than raw, millisecond-level accelerometer time series. While the survey target is denoted `phq9_score_end` (survey administered at period end), exact intra-month survey completion timestamps cannot be independently audited from the Parquet artifacts.

2. **Severe Class Sample Size:** Although the cohort is substantial ($N=4,948$), severe depression (Tier 4) constitutes only 4.6% of test assessments ($N=72$). While threshold tuning improves sensitivity to 41.7%, larger multi-center cohorts are needed to further refine severe classification.

3. **Observational vs. Causal Interpretation:** Permutation and SHAP feature importance reflect associative patterns within the model's decision function. They do not constitute counterfactual causal claims; neither sleep regularization nor physical exercise is causally proven to cure MDD by this observational study.

4. **Absence of Inferential Hypothesis Testing:** While performance differences between sequential models and the persistence baseline are descriptively large (44.6% vs. 0.0% directional agreement), formal non-parametric significance testing (e.g., paired Wilcoxon signed-rank tests) was not conducted across all folds.

5. **Lack of Prospective Clinical Deployment:** This framework is developed and audited strictly for **RESEARCH USE ONLY**. Prospective randomized clinical trials are required before incorporating these algorithms into active clinical workflows.

7. Conclusion and Future Work

This paper presents an explainable machine learning and longitudinal sequence modeling framework for depression severity prediction and progression tracking from passive wearable sensing. By enforcing strict participant-disjoint splitting across 4,948 participants and 10,866 labeled PHQ-9 evaluations, we eliminated identity leakage and established realistic performance benchmarks for real-world affective computing. We resolved the longitudinal Persistence Baseline Paradox, proving that recurrent GRU architectures capture non-zero clinical transitions with up to 44.60% directional agreement where naive baselines fail completely. Furthermore, cost-sensitive threshold optimization ($T_4^* = 0.2100$) increased severe depression recall by 66.7%, and TreeSHAP provided an itemized clinical audit trail attributing 78.9% of predictive power to passive sleep, circadian, and actigraphy biomarkers.

Future work will focus on integrating high-frequency intraday sensor streams, evaluating transfer learning across heterogeneous wearable hardware vendors, implementing multi-task continuous state-space models, and conducting prospective clinical triage trials in outpatient psychiatric clinics.

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